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An atom-photon quantum gate using an atomic clock qubit and a birefringent cavity

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Appendix A

Analytical Derivations

A.1 Analytical derivation for case of atom in $|0\rangle$ and photon is in $|h\rangle$

Our Hamiltonian is as follows:

$$H = \hbar g(|e\rangle \langle 1| a_h + |1\rangle \langle e| a_h^\dagger)$$

We drive the a_h mode of our cavity through $a_h^{in}(t)$, thus giving us the following equation of motion for our a_h cavity mode:

$$\dot{a}_h = -i[a_h, H] - (i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

Because a_h commutes with H , the commutator is 0 and we get

$$\rightarrow \dot{a}_h = -(i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

We now drop the subscript h as we'll only be looking at that case.

Multiplying with $e^{i\omega t}$, and integrating we get:

$$\int dt e^{i\omega t} \dot{a}(t) = \underbrace{[e^{i\omega t} a(t)]_{-\infty}^{\infty}}_{=0} - \underbrace{i\omega \int dt e^{i\omega t} a(t)}_{=i\omega a(\omega)}$$

where in the last expression we used that the integral performs a Fourier transformation.

$$\begin{aligned} \Rightarrow -i\omega a(\omega) &= -(i\Delta + \frac{\kappa}{2})a(\omega) - \sqrt{\kappa}a^{in}(\omega) \\ \sqrt{\kappa}a^{in}(\omega) &= -(i\Delta + \frac{\kappa}{2} - i\omega)a(\omega) \\ \rightarrow a(\omega) &= -\frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega}a^{in}(\omega) \end{aligned}$$

Now from our regular input output relation

$$a^{out}(\omega) = a^{in}(\omega) + \kappa a(\omega)$$

We thus get:

$$\begin{aligned}
a^{out}(\omega) &= a^{in}(\omega) - \frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega} a^{in}(\omega) \\
a^{out}(\omega) &= \left(1 - \frac{\kappa}{i\Delta + \frac{\kappa}{2} - i\omega}\right) a^{in}(\omega) \\
a^{out}(\omega) &= \left(1 - \frac{i\Delta + \frac{\kappa}{2} - i\omega - \kappa}{i\Delta + \frac{\kappa}{2} - i\omega}\right) a^{in}(\omega) \\
a^{out}(\omega) &= \left(\frac{i\Delta - \frac{\kappa}{2} - i\omega}{i\Delta + \frac{\kappa}{2} - i\omega}\right) a^{in}(\omega) \\
&= \underbrace{\frac{i(\Delta - \omega) - \frac{\kappa}{2}}{i(\Delta - \omega) + \frac{\kappa}{2}}}_{h(\omega)} a^{in}(\omega)
\end{aligned}$$

We can then use the convolution theorem in order to solve for $a^{out}(t)$

$$a^{out}(t) = \int_{-\infty}^{\infty} d\tau \underbrace{h(t-\tau)}_{h(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega t} h(\omega)} a^{in}(\tau)$$

Now using that $h(\omega) = 1 - \frac{\kappa}{i(\Delta - \omega) + \frac{\kappa}{2}}$ we get for $h(t)$ (via partial integration):

$$\begin{aligned}
h(t) &= \delta(t) - \kappa e^{-i\delta t} e^{-\frac{\kappa}{2}t} \theta(t) \\
\Rightarrow a^{out}(t) &= a^{in}(t) - \kappa \int_{-\infty}^{\infty} d\tau e^{-i\delta(t-\tau)} e^{-\frac{\kappa}{2}(t-\tau)} a^{in}(\tau)
\end{aligned}$$

For slowly varying pulses we can now approximate that $a^{in}(\tau) \approx a^{in}(t)$ thus giving us:

$$\begin{aligned}
\Rightarrow a^{out}(t) &\approx a^{in}(t) - \frac{\kappa}{i\Delta + \frac{\kappa}{2}} a^{in}(t) \\
&= \frac{i\Delta - \frac{\kappa}{2}}{i\Delta + \frac{\kappa}{2}} a^{in}(t)
\end{aligned}$$

Which for no detuning (so $\Delta = 0$) gives us:

$$a^{out}(t) = -a^{in}(t) \tag{A.1}$$

A.1.1 Lindblad Formalism

We start from the master equation:

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right), \tag{A.2}$$

Now, for an operator O ,

$$\frac{d}{dt} \langle O \rangle = \text{Tr} (O \dot{\rho}) \tag{A.3}$$

$$= \text{Tr} \left(O \left[-\frac{i}{\hbar} [H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right) \right] \right). \tag{A.4}$$

The Hamiltonian contribution gives

$$-\text{Tr} \left(O \frac{i}{\hbar} [H, \rho] \right) = -\frac{i}{\hbar} \text{Tr} (OH\rho - O\rho H) = -\frac{i}{\hbar} \text{Tr} ([O, H]\rho) = -\frac{i}{\hbar} \langle [O, H] \rangle. \quad (\text{A.5})$$

For the dissipative terms we obtain

$$\sum_j \gamma_j \text{Tr} \left(OL_j \rho L_j^\dagger - \frac{1}{2} OL_j^\dagger L_j \rho - \frac{1}{2} O \rho L_j^\dagger L_j \right). \quad (\text{A.6})$$

Using cyclicity of the trace:

$$= \sum_j \gamma_j \left(\langle L_j^\dagger O L_j \rangle - \frac{1}{2} \langle \{L_j^\dagger L_j, O\} \rangle \right). \quad (\text{A.7})$$

Therefore the general equation of motion is

$$\frac{d}{dt} \langle O \rangle = -\frac{i}{\hbar} \langle [O, H] \rangle + \sum_j \gamma_j \left(\langle L_j^\dagger O L_j \rangle - \frac{1}{2} \langle \{L_j^\dagger L_j, O\} \rangle \right). \quad (\text{A.8})$$

A.1.2 Calculation of the Maxwell Bloch Equations (no lossy channels)

A.1.2.1 Jaynes Cummings Hamiltonian

Assuming ground state has zero energy our Jaynes-Cummings Hamiltonian is equal to:

$$H_{JC,S} = \underbrace{\omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma}_{=H_{0,S}} + \underbrace{g(\sigma + \sigma^\dagger)(a + a^\dagger)}_{=H_{int,S}} + H_{drive,S} \quad (\text{A.9})$$

where σ is the atomic lowering operator, ω_a the atomic transition frequency between the levels $|1\rangle$ and $|e\rangle$, a the photonic destruction operator, and ω_a the cavity mode resonance frequency.

H_{drive} in this case is the external light field which is driving our system, which in the semiclassical formalism has the following form [1]:

$$H_{drive,S} = i(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(a + a^\dagger) \quad (\text{A.10})$$

Here, a again is the photonic destruction operator, ω_d is the frequency of the driving field, and ϵ is the complex amplitude of our incoming pulse.

In general going from the Schrödinger picture to the interaction picture, is achieved by converting our operators into the interaction picture via following equation:

$$i \frac{dA_I}{dt} = [A_I(t), H_{0,S}] \quad (\text{A.11})$$

Furthermore, $H_{0,I} = H_{0,S}$. Equipped with this we can now calculate the expressions for operators in the interaction picture:

$$i \frac{da_I(t)}{dt} = [a_I(t), H_{0,S}] \quad (\text{A.12})$$

$$= [a_I(t), \omega_c a^\dagger a] \quad (\text{A.13})$$

$$= \omega_c a_I(t) \quad (\text{A.14})$$

$$\Rightarrow \frac{da_I(t)}{dt} = -i\omega_c a_I(t) \rightarrow a_I(t) = a e^{-i\omega_c t} \quad (\text{A.15})$$

Where we used that:

$$[a, H_{0,S}] = \omega_c [a, a^\dagger a] = \omega_c \underbrace{([a, a^\dagger]) a}_{=1} + a^\dagger \underbrace{[a, a]}_{=0} = \omega_c a \quad (\text{A.16})$$

Now we repeat the same thing for the atomic lowering operator σ :

$$i \frac{d\sigma_I(t)}{dt} = \underbrace{[\sigma_I(t), H_{0,S}]}_{=0} \quad (\text{A.17})$$

$$i \frac{d\sigma_I(t)}{dt} = \omega_a \sigma_I(t) \quad (\text{A.18})$$

$$\Rightarrow \frac{d\sigma_I(t)}{dt} = -i\omega_a \sigma_I(t) \rightarrow \sigma_I(t) = \sigma e^{-i\omega_a t} \quad (\text{A.19})$$

Now we can calculate the Hamiltonian for the interaction part and the drive part, in the interaction picture, by just plugging these operators in there, giving us:

$$H_{int,I} = \sigma a e^{-i(\omega_a - \omega_c)t} + \sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t} + \sigma^\dagger a^\dagger e^{i(\omega_c + \omega_a)t} \quad (\text{A.20})$$

$$\xrightarrow{RWA} = \sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t} \quad (\text{A.21})$$

$$H_{drive,I} = i(\epsilon e^{-i\omega_d t} + \epsilon^\dagger e^{i\omega_d t})(a_I(t) + a_I^\dagger(t)) \quad (\text{A.22})$$

$$\xrightarrow{RWA} = i(a\epsilon^* e^{-i(\omega_c - \omega_d)t} + a^\dagger \epsilon e^{i(\omega_c - \omega_d)t}) \quad (\text{A.23})$$

Thus giving us the total Hamiltonian in the interaction picture under the rotating wave approximation:

$$H_{JC,I} \stackrel{RWA}{=} \omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma + g(\sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t}) + i(a\epsilon^* e^{-i(\omega_c - \omega_d)t} + a^\dagger \epsilon e^{i(\omega_c - \omega_d)t}) \quad (\text{A.24})$$

If we now transform back into the Schrödinger picture, we get:

$$H_{JC,S} \stackrel{RWA}{=} \omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma + g(a^\dagger \sigma + a \sigma^\dagger) + i(a\epsilon^* e^{i\omega_d t} + a^\dagger \epsilon e^{-i\omega_d t}) \quad (\text{A.25})$$

If we now change into the rotating frame of the driving field via

$$U(t) = \exp\{i\omega_d t(a^\dagger a + \sigma^\dagger \sigma)\} \quad (\text{A.26})$$

$$H \rightarrow U H U^\dagger + i\dot{U} U^\dagger =: H_{RF} \quad (\text{A.27})$$

we then get the final form of our Hamiltonian:

$$H_{JC,RF} = \underbrace{(\omega_c - \omega_d)}_{=\Delta_c} a^\dagger a + \underbrace{(\omega_a - \omega_d)}_{=\Delta_a} \sigma^\dagger \sigma + g(a^\dagger \sigma + a \sigma^\dagger) + i(a\epsilon^* + a^\dagger \epsilon) \quad (\text{A.28})$$

Which gives rise to 4 scenarios which I'll be covering with my project:

scenario	meaning
$\Delta_a = \Delta_c = 0$	Atom is in coupled state (here: $ 1\rangle$) and we drive with a light field that's resonant to the cavity (here: we drive $ 1\rangle \leftrightarrow e\rangle$ which would be our $ \pi\rangle$ polarized light)
$\Delta_a = 0; \Delta_c \neq 0$	Atom is in coupled state but we drive with light that's non-resonant to cavity (here: $ V\rangle$ polarized light)
$\Delta_a \neq 0; \Delta_c = 0$	Atom is in uncoupled state (here: $ 0\rangle$) and we drive with a light field that's resonant to the cavity
$\Delta_a, \Delta_c \neq 0$	Atom is in uncoupled state and we drive with light that's non-resonant to cavity

In order to improve readability we'll be splitting the full JC Hamiltonian into three parts:

$$H_0 = \Delta_c a^\dagger a + \Delta_a \sigma^\dagger \sigma \quad (\text{A.29})$$

$$H_{int} = g(a^\dagger \sigma + a \sigma^\dagger) \quad (\text{A.30})$$

$$H_{drive} = i(a\epsilon^* + a^\dagger \epsilon) \quad (\text{A.31})$$

with following losses:

$$\gamma_1 = \gamma, \quad L_1 = \sigma, \quad (\text{A.32})$$

$$\gamma_2 = \kappa, \quad L_2 = a. \quad (\text{A.33})$$

A.1.3 Cavity mode a

For H_0 :

$$[a, H_0] = \Delta_c[a, a^\dagger a] + \Delta_c a \underbrace{[a, \sigma^\dagger \sigma]}_{=0} \quad (\text{A.34})$$

$$= \Delta_c(\underbrace{[a, a^\dagger]}_{=1} a + a^\dagger \underbrace{[a, a]}_{=0}) \quad (\text{A.35})$$

$$= \Delta_c a \quad (\text{A.36})$$

$$\Rightarrow -i\langle[a, H_0]\rangle = -i\Delta_c\langle a \rangle \quad (\text{A.37})$$

For H_{int} :

$$[a, H_{int}] = g([a, a^\dagger \sigma] + [a, a \sigma^\dagger]) \quad (\text{A.38})$$

$$= g(aa^\dagger \sigma - a^\dagger a \sigma + \cancel{aa\sigma^\dagger} - \cancel{aa\sigma^\dagger}) \quad (\text{A.39})$$

$$\rightarrow \langle aa^\dagger \sigma \rangle = \langle n | aa^\dagger \sigma | n \rangle = (n+1)\langle \sigma \rangle \quad (\text{A.40})$$

$$\rightarrow \langle a^\dagger a \sigma \rangle = \langle n | a^\dagger a \sigma | n \rangle = n\langle \sigma \rangle \quad (\text{A.41})$$

$$\Rightarrow -i\langle[a, H_{int}]\rangle = -ig\langle \sigma \rangle \quad (\text{A.42})$$

For H_{drive} :

$$[a, H_{drive}] = [a, i(a\epsilon^* + a^\dagger \epsilon)] \quad (\text{A.43})$$

$$= i([a, a\epsilon^*] + [a, a^\dagger \epsilon]) \quad (\text{A.44})$$

$$= i(0 + \epsilon) \quad (\text{A.45})$$

$$= i\epsilon \quad (\text{A.46})$$

$$\Rightarrow -i\langle[a, H_{drive}]\rangle = i\langle \epsilon \rangle \stackrel{\epsilon \in \mathbb{C}}{=} i\epsilon \quad (\text{A.47})$$

If we now include the fact that:

$$i\epsilon \hat{=} \sqrt{\kappa_{oc}} a^{in}(t) \quad (\text{A.48})$$

we are then left with:

$$\Rightarrow -i\langle[a, H_{drive}]\rangle = -\sqrt{\kappa_{oc}} a^{in}(t) \quad (\text{A.49})$$

For rest of the Lindbladian:

$$D(a) = a^\dagger aa - \frac{1}{2}\{a^\dagger a, a\} \quad (\text{A.50})$$

$$= a^\dagger aa - \frac{1}{2}(a^\dagger aa - aa^\dagger a) \quad (\text{A.51})$$

$$= \frac{1}{2}a^\dagger aa - \frac{1}{2}aa^\dagger a \quad (\text{A.52})$$

$$= \frac{1}{2}(a^\dagger aa - aa^\dagger a) \quad (\text{A.53})$$

$$= \frac{1}{2}([a^\dagger, a]a) \quad (\text{A.54})$$

$$= \frac{1}{2}(-a) \quad (\text{A.55})$$

$$= -\frac{1}{2}a \quad (\text{A.56})$$

$$\Rightarrow \kappa\langle D(a) \rangle = -\frac{\kappa}{2}\langle a \rangle \quad (\text{A.57})$$

So for $\langle a \rangle(t)$ we get:

$$\begin{aligned} \frac{d\langle a \rangle}{dt} &= -i\Delta_c \langle a \rangle - ig\langle \sigma \rangle - \sqrt{\kappa_{oc}} a_{in}(t) - \frac{\kappa}{2} \langle a \rangle \\ &= -(i\Delta_c + \frac{\kappa}{2}) \langle a \rangle - ig\langle \sigma \rangle - \sqrt{\kappa_{oc}} a_{in}(t) \end{aligned} \quad (\text{A.58})$$

A.1.4 Atomic lowering operator σ

For H_0 :

$$[\sigma, H_0] = \Delta_a \underbrace{[\sigma, \sigma^\dagger \sigma]}_{=1} \quad (\text{A.59})$$

$$\begin{aligned} &= \underbrace{[\sigma, \sigma^\dagger]}_{=1} \sigma + \sigma^\dagger \underbrace{[\sigma, \sigma]}_{=0} \\ &= \Delta_a \mathbb{1} \sigma \end{aligned} \quad (\text{A.60})$$

$$\Rightarrow -i\langle [\sigma, H_0] \rangle = -i\Delta_a \langle \sigma \rangle \quad (\text{A.61})$$

For H_{int} :

$$[\sigma, H_{int}] = g([\sigma, a^\dagger \sigma] + [\sigma, a \sigma^\dagger]) \quad (\text{A.62})$$

$$= g(\cancel{a^\dagger \sigma \sigma} - \cancel{a^\dagger \sigma \sigma} + a \sigma \sigma^\dagger - a \sigma^\dagger \sigma) \quad (\text{A.63})$$

$$= g(a \sigma \sigma^\dagger - a \sigma^\dagger \sigma) \quad (\text{A.64})$$

Now we can consider that:

$$\sigma^\dagger \sigma = |e\rangle \langle 1| \langle e| = |e\rangle \langle e| \quad (\text{A.65})$$

$$\sigma \sigma^\dagger = |1\rangle \langle e| \langle 1| = |1\rangle \langle 1| \quad (\text{A.66})$$

$$[\sigma, H_{int}] = g(a \sigma \sigma^\dagger - a \sigma^\dagger \sigma) \quad (\text{A.67})$$

$$= ga(|1\rangle \langle 1| - |e\rangle \langle e|) \quad (\text{A.68})$$

$$= ga\sigma_z \quad (\text{A.69})$$

$$\Rightarrow -i\langle [\sigma, H_{int}] \rangle = -ig\langle a\sigma_z \rangle \quad (\text{A.70})$$

No interaction with H_{drive} .

For rest of Lindbladian:

$$D(\sigma) = \sigma^\dagger \underbrace{\sigma \sigma}_{=0} - \frac{1}{2} \{ \sigma^\dagger \sigma, \sigma \} \quad (\text{A.71})$$

$$= -\frac{1}{2} \underbrace{\sigma^\dagger \sigma \sigma}_{=0} - \frac{1}{2} \sigma \sigma^\dagger \sigma \quad (\text{A.72})$$

$$= -\frac{1}{2} \sigma \quad (\text{A.73})$$

$$\gamma \langle D(\sigma) \rangle = -\frac{\gamma}{2} \langle \sigma \rangle \quad (\text{A.74})$$

So for $\langle \sigma \rangle(t)$ we get:

$$\begin{aligned} \frac{d\langle \sigma \rangle}{dt} &= -i\Delta_a \langle \sigma \rangle - ig\langle a\sigma_z \rangle - \frac{\gamma}{2} \langle \sigma \rangle \\ &= -(i\Delta_a + \frac{\gamma}{2}) \langle \sigma \rangle - ig\langle a\sigma_z \rangle \end{aligned} \quad (\text{A.75})$$

A.1.5 Pauli-Z operator σ_z

For H_0 :

$$[\sigma_z, H_0] = \Delta_a [\sigma_z, \sigma^\dagger \sigma] \quad (\text{A.76})$$

$$= \Delta_a [\sigma_z, |e\rangle \langle e|] \quad (\text{A.77})$$

$$= \Delta_a (\sigma_z |e\rangle \langle e| - |e\rangle \langle e| \sigma_z) \quad (\text{A.78})$$

$$= \Delta_a (-|e\rangle \langle e| + |e\rangle \langle e|) \quad (\text{A.79})$$

$$= 0 \quad (\text{A.80})$$

For H_{int} :

$$[\sigma_z, H_{int}] = g([\sigma_z, a^\dagger \sigma] + [\sigma_z, a \sigma^\dagger]) \quad (\text{A.81})$$

$$= g(a^\dagger \sigma_z \sigma - a^\dagger \sigma \sigma_z + a \sigma_z \sigma^\dagger - a \sigma^\dagger \sigma_z) \quad (\text{A.82})$$

$$= g(a^\dagger \sigma + a^\dagger \sigma - a \sigma^\dagger - a \sigma^\dagger) \quad (\text{A.83})$$

$$= 2g(a^\dagger \sigma - a \sigma^\dagger) \quad (\text{A.84})$$

$$\boxed{\Rightarrow -i\langle [\sigma_z, H_{int}] \rangle = -2ig(\langle a^\dagger \sigma \rangle - \langle a \sigma^\dagger \rangle)} \quad (\text{A.85})$$

No interaction with H_{drive} . For rest of Lindbladian:

$$D(\sigma_z) = \sigma^\dagger \sigma_z \sigma - \frac{1}{2} \{ \sigma^\dagger \sigma, \sigma \} \quad (\text{A.86})$$

$$= |e\rangle \langle e| - \frac{1}{2} \underbrace{\sigma^\dagger \sigma \sigma_z}_{=-|e\rangle \langle e|} - \frac{1}{2} \underbrace{\sigma_z \sigma^\dagger \sigma}_{=-|e\rangle \langle e|} \quad (\text{A.87})$$

$$= |e\rangle \langle e| + |e\rangle \langle e| \quad (\text{A.88})$$

$$= 2|e\rangle \langle e| \quad (\text{A.89})$$

$$= \mathbb{1} - \sigma_z \quad (\text{A.90})$$

$$\boxed{\Rightarrow \gamma \langle D(\sigma_z) \rangle = \gamma(1 - \langle \sigma_z \rangle)} \quad (\text{A.91})$$

So for $\langle \sigma_z \rangle(t)$ we get:

$$\boxed{\frac{d\langle \sigma_z \rangle(t)}{dt} = -2ig(\langle a^\dagger \sigma \rangle - \langle a \sigma^\dagger \rangle) + \gamma(1 - \langle \sigma_z \rangle)} \quad (\text{A.92})$$

The final Maxwell Bloch Equations then take the form of:

$$\boxed{\frac{d\langle a \rangle}{dt} = -(i\Delta_c + \frac{\kappa}{2})\langle a \rangle - ig\langle \sigma \rangle - \sqrt{\kappa_{oc}} a_{in}(t)} \quad (\text{A.93})$$

$$\boxed{\frac{d\langle \sigma \rangle}{dt} = -(i\Delta_a + \frac{\gamma}{2})\langle \sigma \rangle - ig\langle a \sigma_z \rangle} \quad (\text{A.94})$$

$$\boxed{\frac{d\langle \sigma_z \rangle}{dt} = -2ig(\langle a^\dagger \sigma \rangle - \langle a \sigma^\dagger \rangle) + \gamma(1 - \langle \sigma_z \rangle)} \quad (\text{A.95})$$

Bibliography

- [1] D. F. Walls and G. J. Milburn. *Quantum Optics*. 2nd. Berlin, Heidelberg: Springer, 2008. ISBN: 978-3-540-28574-8. DOI: 10.1007/978-3-540-28574-8.

Selbständigkeitserklärung

Ich versichere hiermit, dass ich die von mir eingereichte Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

gezeichnet,
Leart Zuka, München den TBD.